GA 11 – Task 1 Engineering Project and Business plan

COURSE: Diploma in Electrical Engineering

MODULE: Engineering Design 2

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PHASE 1 - PROPOSAL

The engineering problem is that small organisations such as school labs and small workshops need a more affordable way of distributing power to their projects. Commercial controllers are expensive and when they are damaged there is no way of fixing the device, instead a new unit would have be purchased which results in more costs being generated.

The client is a local technical school that builds small DC projects. They require an analogue adjustable PCB that is simple to use, cost-effective, and easy to repair. It requires parts that are easily accessible.

The project objectives are to design a schematic, manufacture a PCB and test the prototype to ensure the client's specifications are met. These objectives must be completed within the allocated timeframe and budget.

The proposed solution is a compact analogue control board that is based on an adjustable feedback circuit with an LM324N operational amplifier circuit. The 1N4733A Zener diode is used as a reference voltage and the 2N4401TA and TIP29AG transistors pair supply the load, a potentiometer allows the user to tune the output. Terminal blocks connect the supply and the load, and an LED is used to display circuit operation.

PHASE 2 – BUDGET PLANNING

Component	**Quantity**	**Estimated Unit Cost**	**Estimated Total**
LM324N IC	1	R15	R15
2N4401TA NPN transistor	1	R8	R8
TIP29AG NPN power transistor	1	R15	R15
Resistor	10	R2	R20
PT10LV10-201A2020 potentiometer	2	R15	R30
120 µF electrolytic capacitor	2	R5	R10
1N4148-1 diode	4	R2	R8
Standard LED	2	R3	R6
1N4773A Zener diode	1	R4	R4
Terminal Block 01X02P-5.00 mm	2	R10	R20
Single-sided PCB	1	R60	R60
Solder, wires, and small consumables	1	R40	R40
3D enclosure box and manufacturing PCB CAD Software (Easy EDA) Station for soldering	1 1 1	R150 R0 R0	R150 R0 R0
Subtotal			R386
Contingency for damaged/replaced components			R40
Estimated Project Budget			R426

The proposed budget for our analogue adjustable PCB shows that the project can cost-effectively be completed and with assistance from equipment such as soldering stations, multi meters and other laboratory equipment being readily available to use at the university upon permission being granted. The largest project costs are from the electrical hardware of

the product while designing tools for the circuit and PCB such as “Easy EDA” allows for the designing of the components online.

PHASE 3 – RISK ANALYSIS

Risk	Category	Likelihood	Impact	Mitigation Strategy
Required components unavailable	Technical	Medium	High	Identify alternative suppliers before placing the order.
Incorrect footprint or component dimensions	Technical	Medium	Medium	Verify all footprints against the datasheet before finalizing.
PCB routing errors	Technical	Medium	High	Do design-rule checks and do proper inspection of tracks and connections before manufacturing.
Damage of components during soldering	Technical	Medium	High	Practice soldering beforehand and follow the correct soldering procedures.
Total project cost exceeds the allocated budget	Financial	Low	High	Keep track of spending against the budget after every purchase; substitute non-critical parts with cheaper generics if needed.
Product does not perform as it should during testing	Safety	Medium	High	Test while building during each stage to ensure operation. Check for soldering faults throughout building.

PHASE 4 – TIME AND CHANGE MANAGEMENT

Activity	Planned duration	Milestone
Study circuit and establish requirements	1 Week	Requirement agreed
Capture of schematic in the CAD software	1 Week	Schematic finished
Placement of components on PCB CAD	1 Week	PCB layout completed
Design-rule checks and making of corrections	1 Week	Correction of PCB completed
Component collection	3 Weeks	All components available
Assembling of PCB and board enclosure physically	2 Weeks	Assembly completed
Testing and checking of faults	1 Week	Fault-checking completed

Demonstration of product and providing evidence	1 Week	Project completed

APPROVED PROJECT CHANGE

Some of the original components from what we were given by the lecture were not available from the software that we were using and was replaced with equivalent parts with the same rating. The substitution that was made was reviewed by the team to ensure that electrical functionality and compatibility was maintained. The change was recorded and did not require an alteration to the project objectives.

PHASE 5 – QUALITY ASSURANCE REPORT

Requirement	Verification method	Result
Circuit operates from specified 20 V DC Supply	Connect the circuit to the power supply and administer operation for a duration of 8 hours	✓ Pass
Output voltage is adjustable	Monitor the circuit operation of the potentiometers and measure voltage	✓ Pass
LED provides visual indication of operation	Supply power to the circuit and monitor LED operation for 8 hours	✓ Pass
Components are correctly mounted and follow the correct circuit operation	Inspect and verify that circuit connections are the same as schematic and PCB layout	✓ Pass
Check for no soldering joints and that proper soldering procedures were followed	Inspect the bottom of the board for any soldering mistakes and fix any if they are found	✓ Pass
Terminal connections visibly identifiable	Visually inspect that the correct terminals are clearly identified	✓ Pass
Output voltage and output current are measurable	Measure the currents and voltages using the multi meter	✓ Pass
Operation is safe	The circuit operation was repeatedly monitored	✓ Pass
Enclosure compact and fits the circuit board and wires	Check that wires and the PCB are well spaced inside the box	✓ Pass

SUMMARY

The completed prototype satisfied all the requirements that the client specified, and safe operation was monitored during testing. The analogue adjustable PCB worked as planned due to power being well-supplied to the circuit, the circuit being adjustable and relatively simple to use. The components were well-fitted into the box in a manner that allowed for proper circuit operation.

PHASE 6 – ENGINEERING ECONOMICS

INITIAL COST

The estimated cost of the analogue adjustable PCB is R426 which includes electrical components, the single-sided PCB, the 3D box enclosure, consumables and contingency.

LONG-TERM VALUE

The finalized product is adjustable as it allows for modification of the output without making changes to the PCB circuit layout. The PCB is housed in a sealed enclosure with a removable lid, making it protected and safe in the field and easy to open if recalibrating or repairing is ever required. The use of replaceable electrical components promotes efficient maintenance as opposed to replacing the entire unit.

ECONOMIC EVALUATION

The project costs are estimated to be R426 which is relatively affordable. Especially since free software is available, the PCB requires standard electrical components and electrical equipment, so the costs are reduced. The design provides potential long term use due to its being adjustable, reuseable, and repairable.

RECOMMENDATIONS

The proposed engineering solution is recommended due to the costs involved being a bargain for all the benefits that the product offers. The long-term value of the PCB outweighs the short-term costs by a great margin.

PHASE 7 – EVALUATION OF PROJECT OUTCOMES

PROJECT OUTCOME

The project resulted in a successful analogue adjustable PCB which produced functional operation during testing. The completed project was evaluated alongside the design of the PCB circuit schematic, the PCB construction and layout and verified to have met the client's requirements which included adjustable operation, reusability, and simple functionality. The final output voltage and current that was measured during testing confirmed circuit performance. Overall, the project object was achieved as the prototype functioned in the manner in which it was planned.

STRENGTHS

- The output voltage was adjustable through the potentiometer feedback network.
- The LED provided clear indication of circuit operation.
- The PCB being analogue based further made the project simple to build as no programming was required.
- The costs involved to build the prototype are affordable and manageable.
- The project required equipment that is readily available and not a hassle to access.

Limitations

- Components availability using the selected CAD software required for equivalent substitution of components.
- Manual checking of terminals is used so first-time users require a set of instructions to operate the PCB.
- During soldering inspection faults were made that needed to be fixed.

Recommendations

- In the future, the project should add clearer marking for calibration to allow for first time or non-technical users to easily make output adjustments.
- The circuit should be planned using exact components that are readily available to eliminate the need for substitutions.
- Components should be sourced directly from suppliers earlier in the project to reduce delays from unavailable parts.
- An improved version should include remote control instead of having to manually operate the circuit.